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ICT – 101

Module 5 Unit 1 Journal

1. Discuss the 3 pillars of Information Assurance in relation to safeguarding information or data

Confidentiality - It guarantees that only authorized people can access information. This includes protecting the information from illegal access, disclosure, or theft. It is essential for protecting delicate information, such as private information, financial information, or confidential data.

Integrity – assures the truthfulness, precision, and integrity of the information. In other words, data must be safeguarded against illegal change, deletion, or corruption. It is crucial for preserving the accuracy and credibility of information.

Availability - guarantees that data is available and useable when required. This entails safeguarding data against interruptions, outages, and denial-of-service assaults. It is crucial for making sure that authorized users get prompt access to the data they require.

2. Briefly explain the attack methods on how these 3 pillars can be compromised.

Hacking and Data leaks compromises **Confidentiality**, these individuals gain access to sensitive data by exploiting system vulnerabilities or using stolen credentials. Confidential information is accidentally or intentionally released to unauthorized parties.

Malware Infections and Unauthorized Changes compromises **Integrity**, attackers modify, delete, or corrupt data, compromising its accuracy and reliability and Unauthorized individuals make unauthorized modifications to data, compromising its integrity.

Ransomware Attacks and Hardware Failures compromises **Availability**, attackers encrypt data and demand payment for decryption, rendering information inaccessible and Hardware malfunctions or outages can disrupt access to information systems and data.

3. Explain how Authentication Authorization and Accounting contribute to Information assurance

By implementing access control, tracking user behavior, and upholding responsibility, the crucial architecture of authentication, authorization, and accounting (AAA) greatly helps to information assurance (IA).

Authentication, it confirms the user identities and limiting access to those who have been given permission. This procedure promotes availability by ensuring that authorized users may access resources without being hindered, protects integrity by lowering the danger of data tampering, and enhances confidentiality by preventing unauthorized access to sensitive data.

For **Authorization**, it limits the users to only the information they are authorized to view or alter by specifying access privileges. This approach ensures availability by making sure resources are distributed properly, upholds integrity by preventing unauthorized actions, and enforces confidentiality by restricting access to sensitive data.

For **Accounting**, it promotes availability by proactively identifying potential performance issues that could influence system uptime, ensures integrity by detecting unauthorized changes, and enhances confidentiality by enabling the detection of unwanted access attempts.

Organizations may successfully safeguard their information assets, uphold the confidentiality, integrity, and availability of data, and improve overall information assurance by employing **AAA** techniques.

4. Think of an asset that is in your possessions or an application that you use that holds confidential information. What security measures or practices have you done to safeguard your data?

I would've done Data Encryption, use Strong passwords and Access Controls to protect my Data. I work to safeguard the confidentiality, integrity, and accessibility of the information entrusted to me by putting these security measures into place and abiding by best practices.